(6) If $A=\begin{bmatrix}1 & -1\\ 2 & -1\end{bmatrix}$, $B=\begin{bmatrix}a & 1\\ b & -1\end{bmatrix}$ and $(A+B)^2=A^2+B^2$, then the values of a and b are

(1) $a=4, b=1$ (2) $a=1, b=4$ (3) $a=0, b=4$ (4) $a=2, b=4$

(7) If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ a & 2 & b\end{bmatrix}$ is a matrix satisfying the equation $AA^T=9I$, where I is 3 × 3 identity matrix, then the ordered pair (a, b) is equal to

(1) $(2, -1)$ (2) $(-2, 1)$ (3) $(2, 1)$ (4) $(-2, -1)$

(8) If A is a square matrix, then which of the following is not symmetric?

(1) $A+A^T$ (2) AA^T (3) A^TA (4) $A-A^T$

(9) If A and B are symmetric matrices of order n, where $(A\neq B)$, then

(1) $A+B$ is skew-symmetric (2) $A+B$ is symmetric

(3) $A+B$ is a diagonal matrix (4) $A+B$ is a zero matrix

(10) If $A=\begin{bmatrix}a & x\\ y & a\end{bmatrix}$ and if $xy=1$, then det $(A\,A^T)$ is equal to

(1) $(a-1)^2$ (2) $(a^2+1)^2$ (3) a^2-1 (4) $(a^2-1)^2$

(11) The value of x, for which the matrix $A=\begin{bmatrix}e^{x-2} & e^{7+x}\\ e^{2+x} & e^{2x+3}\end{bmatrix}$ is singular

(1) 9 (2) 8 (3) 7 (4) 6

(12) If the points $(x,-2)$, $(5,2)$, $(8,8)$ are collinear, then x is equal to

(1) -3 (2) $\frac{1}{3}$ (3) 1 (4) 3

(13) If $\begin{vmatrix}2a & x_1 & y_1\\ 2b & x_2 & y_2\\ 2c & x_3 & y_3\end{vmatrix}=\frac{abc}{2}\neq 0$, then the area of the triangle whose vertices are $\left(\frac{x_1}{a},\frac{y_1}{a}\right)$, $\left(\frac{x_2}{b},\frac{y_2}{b}\right)$, $\left(\frac{x_3}{c},\frac{y_3}{c}\right)$ is

(1) $\frac{1}{4}$ (2) $\frac{1}{4}abc$ (3) $\frac{1}{8}$ (4) $\frac{1}{8}abc$

(14) If the square of the matrix $\begin{bmatrix}\alpha & \beta\\ \gamma & -\alpha\end{bmatrix}$ is the unit matrix of order 2, then α, β and γ should satisfy the relation.

(1) $1+\alpha^2+\beta\gamma=0$ (2) $1-\alpha^2-\beta\gamma=0$

(3) $1-\alpha^2+\beta\gamma=0$ (4) $1+\alpha^2-\beta\gamma=0$

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